

NLP TUTOR

Sequence-to-Sequence (Seq2Seq) Models

Previous session

- We understood why ANN/CNN structures fail for sequential data.
- We explored the **RNN architecture**.
- Learned how RNNs remember previous inputs while processing the next ones using **recurrent connections**.
- Discussed the **types of RNNs**.

Today's Session Overview

- Numerical Example: Forward Computation in RNN (step-by-step)

Day	Mon	Tue	Wed	Thu	Fri	Mon	Tue	Wed	Thu	Fri	Mon
Items sold	9	7	6	10	8	7	11	9	6	12	?



We need to build the model for such applications where:

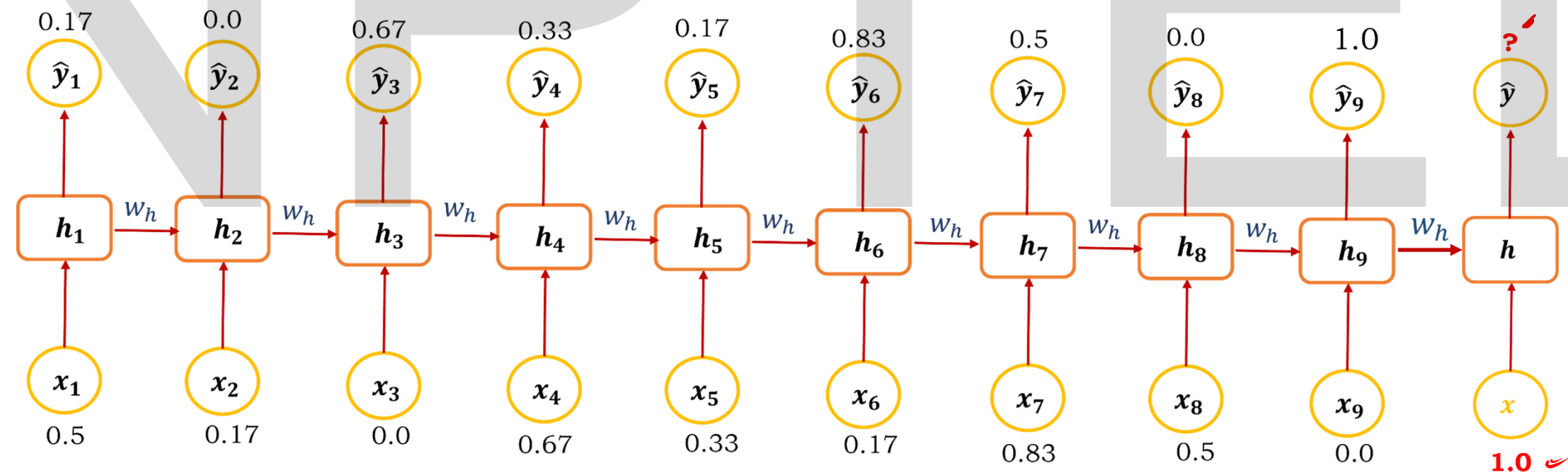
1. The model must **handle dependencies between inputs**
2. The model must **handle variable-length inputs**

To achieve (1) and (2), At each time step, **the function executed by the model should be the same.**

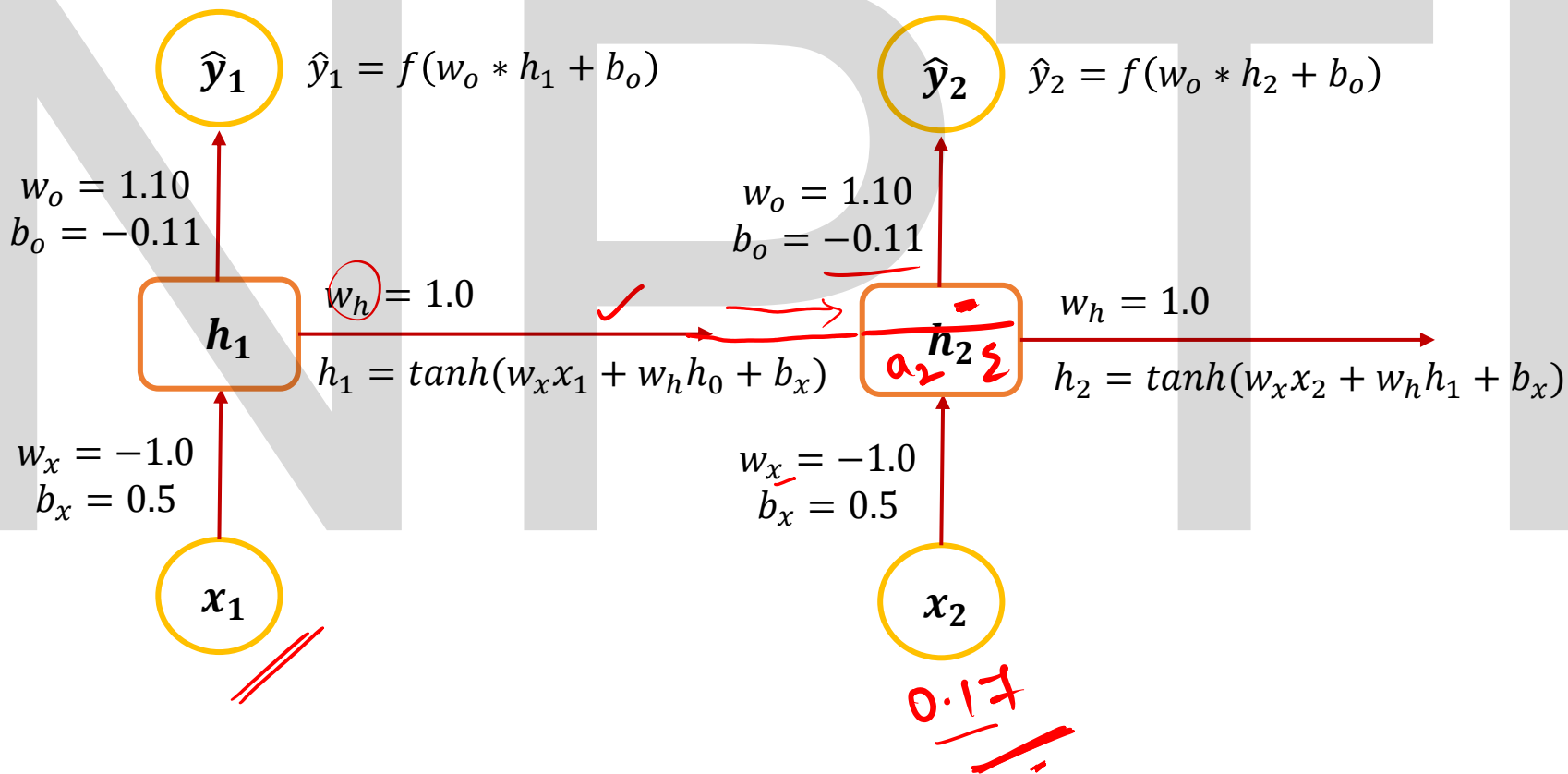
	Mon	Tue	Wed	Thu	Fri	Mon	Tue	Wed	Thu	Fri
	9	7	6	10	8	7	11	9	6	12
X	0.5	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	1.0
Y	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	1.0	

The sales data across several days.
Predict the next value in a time series.

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$



X	0.5	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	
Y (GT)	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	1.0	
$\hat{y}$	-0.11	0.2404								



$$x_2 = 0.17$$

$$a_2 = x_2 w_x + \underbrace{h_1 w_h}_{0.2404} + b_x$$

$$= 0.17(-1.0) + 0(1.0) + 0.5$$

$$a_2 = 0.33$$

$$h_2 = \tanh(a_2) = \tanh(0.33)$$

$$\frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$h_2 \approx 0.3185$$

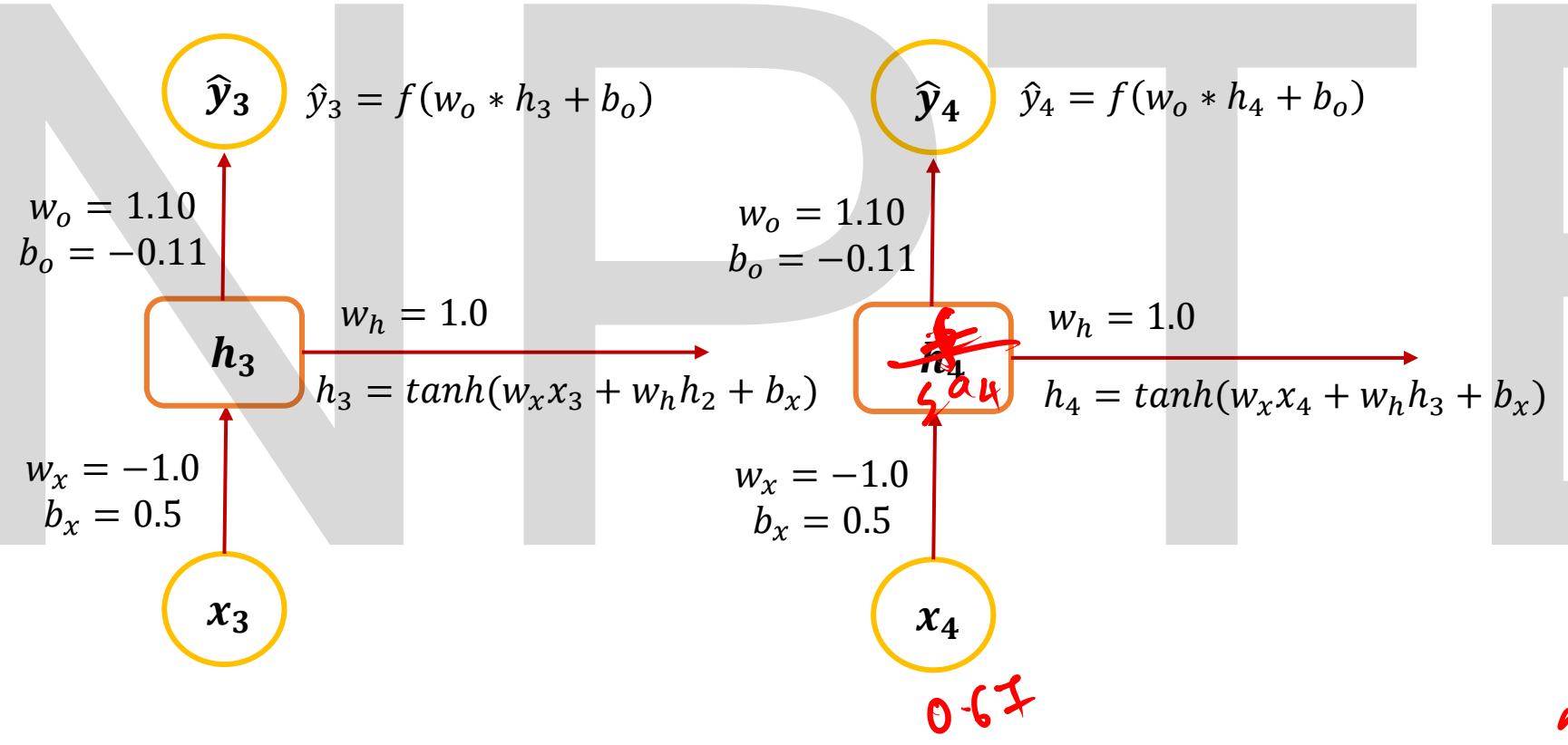
$$\hat{y}_2 = f(h_2 \cdot w_o + b_o)$$

$$\hat{y}_2 = \underline{0.3185(1.10) + (-0.11)}$$

$$\hat{y}_2 = f(0.2404)$$

$$\hat{y}_2 = 0.2404$$

X	0.5	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	
Y (GT)	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	1.0	
$\hat{y}$				<u>0.4020</u>						



Handwritten calculations:

$$x_4 = 0.67$$

$$a_4 = x_4 w_x + w_h \cdot h_3 + b_x$$

$$a_4 = 0.67(-1.0) + 0.6743(1.0) + 0.5$$

$$a_4 \approx 0.5043$$

$$h_4 = \tanh(0.5043)$$

$$h_4 \approx 0.4655$$

$$\hat{y}_4 = f(h_4 \cdot w_o + b_o)$$

$$\hat{y}_4 = \underline{0.4655(1.10) + (-0.11)}$$

$$\hat{y}_4 \approx f(0.4020)$$

$$\hat{y}_4 = 0.4020$$



$$x_6 = 0.17$$

$$a_6 = x_6 \cdot w_x + h_5 w_h + b_x$$

$$a_6 = 0.17(-1.0) + 0.5618(1.0) + 0.5$$

$$a_6 = 0.8918$$

$$h_6 = \tanh(a_6)$$

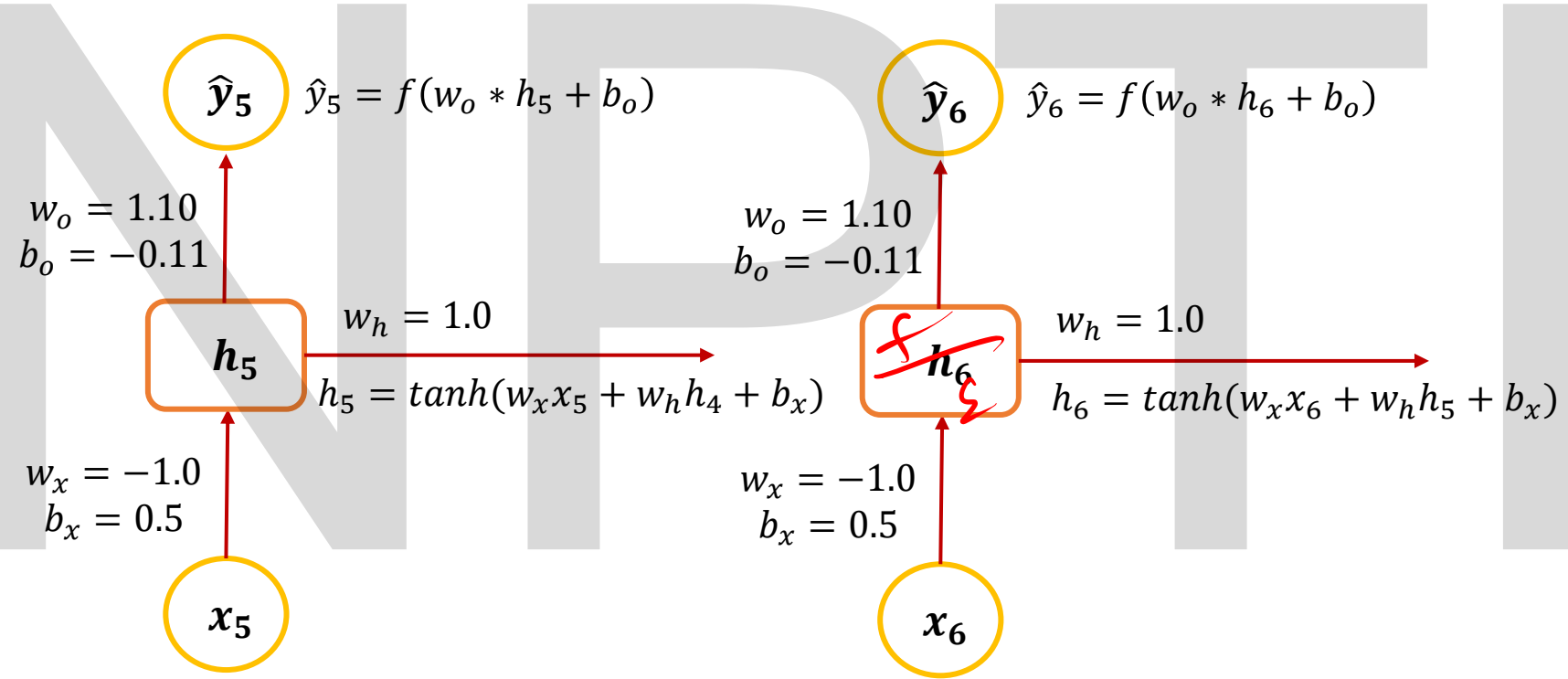
$$h_6 \approx 0.7123$$

$$\hat{y}_6 = f(h_6 \cdot w_o + b_o)$$

$$\hat{y}_6 = f(0.6735)$$

$$\hat{y}_6 = 0.6735$$

X	0.5	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	
Y (GT)	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	1.0	
$\hat{y}$						0.6735				



X	0.5	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	✗
Y (GT)	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	1.0	
$\hat{y}$								0.2743		

$x_8 = 0.50$
 $a_8 = x_8 w_x + h_7 w_h + b_a$
 $a_8 = 0.3647$

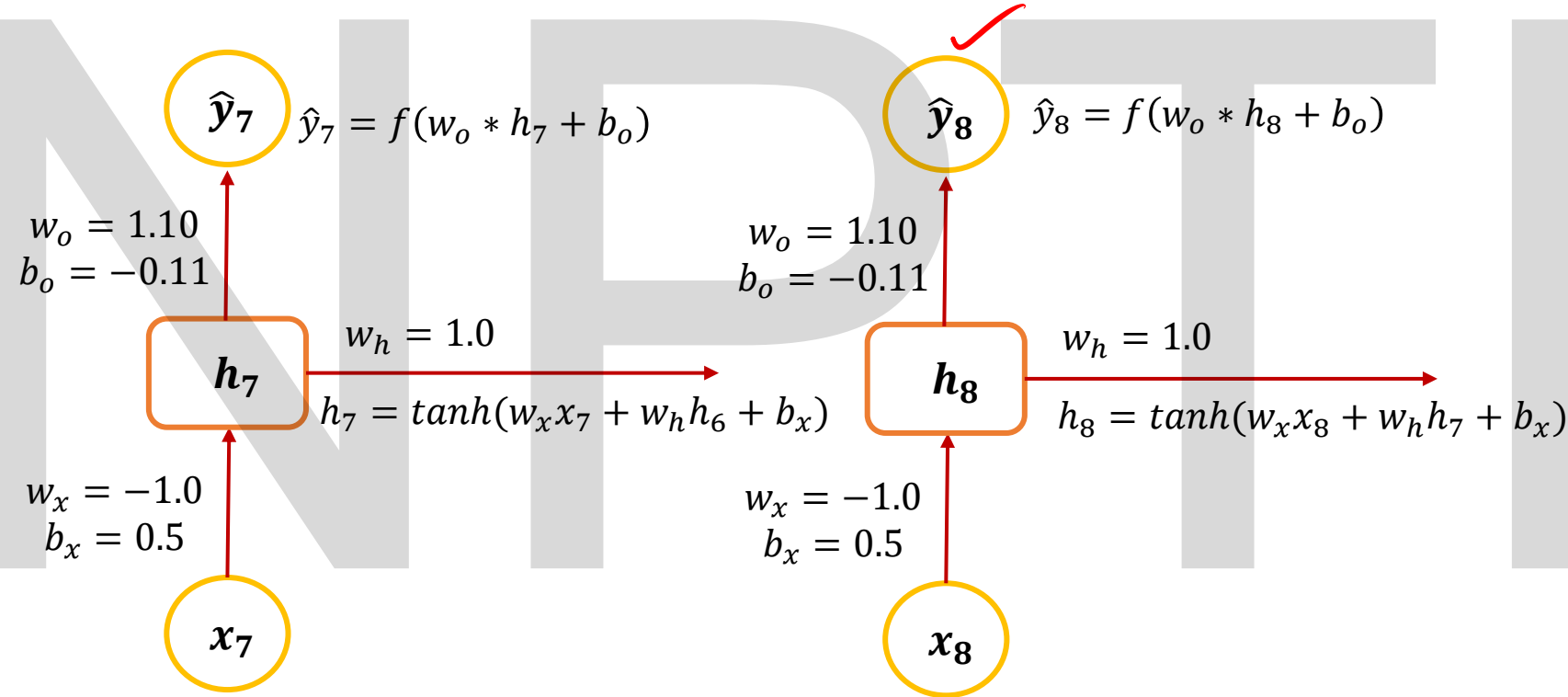
$h_8 = \tanh(a_8)$

$h_8 \approx 0.3493$

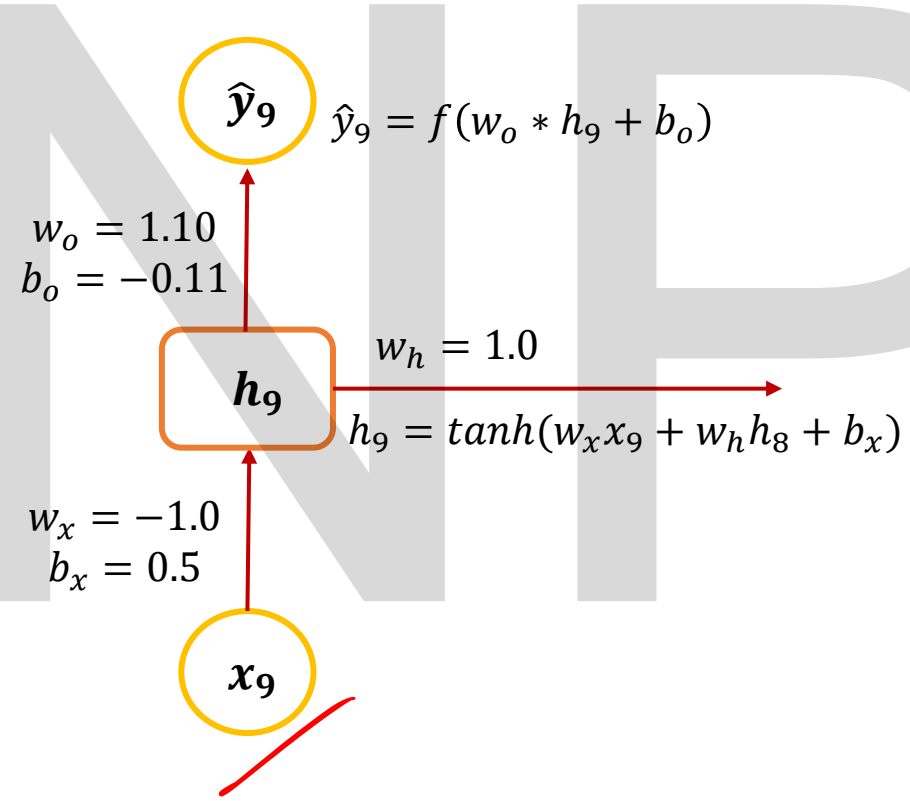
$\hat{y}_8 = f(h_8 \cdot w_o + b_o)$

$\hat{y}_8 = f(0.2743)$

$\hat{y}_8 = 0.2743$



X	0.5	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	✗
Y (GT)	0.17	0.0	0.67	0.33	0.17	0.83	0.5	0.0	1.0	
$\hat{y}$										



w_h
 w_o
 w_x

$x_9 \approx 0.00$

$$a_9 = x_9 w_x + h_8 w_h + b_x$$

$$a_9 = 0.8493$$

$$h_9 = \tanh(a_9)$$

$$h_9 \approx 0.6907$$

$$\hat{y}_9 = f(h_9 \cdot w_o + b_o)$$

$$\hat{y}_9 \approx f(0.6498)$$

$$\hat{y}_9 = 0.6498$$

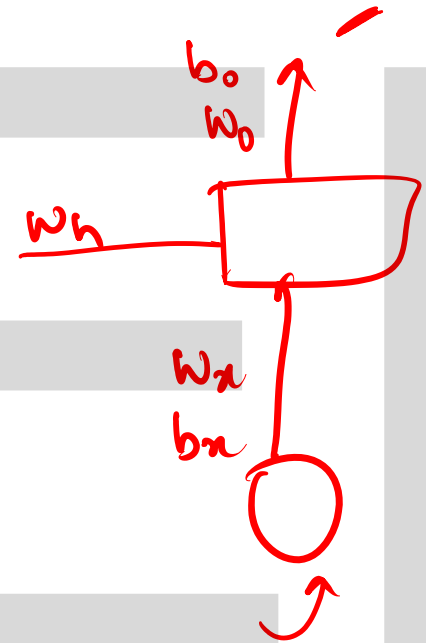
t	x_t	z_t	h_t	$\hat{y}_t$	y_t
1	0.50	0.0000	0.0000	-0.1100	0.17
2	0.17	0.3300	0.3185	0.2404	0.00
3	0.00	0.8185	0.6743	0.6317	0.67
4	0.67	0.5043	0.4655	0.4020	0.33
5	0.33	0.6355	0.5618	0.5080	0.17
6	0.17	0.8918	0.7123	0.6735	0.83
7	0.83	0.3823	0.3647	0.2912	0.5
8	0.50	0.3647	0.3493	0.2743	0.0
9	0.00	0.8493	0.6907	0.6498	1.0

$$MSE = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2$$

$$MSE = \frac{1}{9} ((0.17 - (-0.11))^2 + (0.00 - 0.2402)^2 + \dots + (1.0 - 0.6498)^2)$$

$$MSE = \frac{0.5230}{9} \approx \underline{\underline{0.058}}$$

w_{x^*}
 w_0
 w_h



Next Session

- Backpropagation Through Time (BTT)
- Challenges such as Vanishing and Exploding Gradients

NIPTELL

Thank you